CLOSURES OF POINTS COMMUTE WITH DIRECTED LIMITS OF DECENT SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1215-1247, following the statement at 1205-1213.

Editorial note. No omissions; the statement covers both the set-level and the scheme-structure-level conclusion.

Allowed standard prerequisites (editorial).

- Underlying sets of limits of decent spaces. lemma-inverse-limit-sets (same chapter, card L26292, cited at ~1219-1220). |X| = lim |X_i| as sets.
- Topology of limits. lemma-topology-limit (same chapter, cited at ~1224). |X| = lim |X_i| as topological spaces.
- Describing opens in limits of topological spaces. Topology, Lemma topology-lemma-describe-limits (cross-chapter, cited at ~1225-1227). U = union of preimages of opens in the X_i.
- Reduced induced closed subspace structure. Properties of Spaces, Definition spaces-properties-definition-reduced-induced-space (cross-chapter, cited at ~1233-1235). Endows Z and Z_i with their reduced induced structures.
- Limits of systems of closed immersions. lemma-directed-inverse-system-closed-immersions (same chapter, cited at ~1240-1241). lim Z_i -> X is a closed immersion.
- Limits of reduced spaces are reduced. lemma-directed-inverse-system-reduced (same chapter, cited at ~1242-1243). lim Z_i is reduced.

Source reference. Original lemma 0CUH: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 972–972. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07SF: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 878–878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CUI: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1002–1002. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0CUJ: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1093–1093. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 086V: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1156–1156. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 047X: focused-seeds/geometry/S9-spaces-limits-inverse-limit-irreducibles/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A2P: focused-seeds/geometry/S9-spaces-limits-inverse-limit-irreducibles/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $X = \lim_{i \in I} X_i$ be the limit of a directed inverse system of algebraic spaces over S with affine transition morphisms (Lemma 07SF). Let $x \in |X|$ with images $x_i \in |X_i|$. If each X_i is decent, then $\overline{\{x\}} = \lim_i \overline{\{x_i\}}$ as sets and as algebraic spaces if endowed with reduced induced scheme structure.*

Proof. Set $Z = \overline{\{x\}} \subset |X|$ and $Z_i = \overline{\{x_i\}} \subset |X_i|$. Since $|X| \to |X_i|$ is continuous we see that Z maps into Z_i for each i. Hence we obtain an injective map $Z \to \lim Z_i$ because $|X| = \lim |X_i|$ as sets (Lemma 0CUJ). Suppose that $x' \in |X|$ is not in Z. Then there is an open subset $U \subset |X|$ with $x' \in U$ and $x \notin U$. Since $|X| = \lim |X_i|$ as topological spaces (Lemma 086V) we can write $U = \bigcup_{j \in J} f_j^{-1}(U_j)$ for some subset $J \subset I$ and opens $U_j \subset |X_j|$, see Topology, Lemma 0A2P. Then we see that for some $j \in J$ we have $f_j(x') \in U_j$ and $f_j(x) \notin U_j$. In other words, we see that $f_j(x') \notin Z_j$. Thus $Z = \lim Z_i$ as sets.

Next, endow Z and Z_i with their reduced induced scheme structures, see Properties of Spaces, Definition 047X. The transition morphisms $X_{i'} \to X_i$ induce affine morphisms $Z_{i'} \to Z_i$ and the projections $X \to X_i$ induce compatible morphisms $Z \to Z_i$. Hence we obtain morphisms $Z \to \lim Z_i \to X$ of algebraic spaces. By Lemma 0CUH we see that $\lim Z_i \to X$ is a closed immersion. By Lemma 0CUI the algebraic space $\lim Z_i$ is reduced. By the above $Z \to \lim Z_i$ is bijective on points. By uniqueness of the reduced induced closed subscheme structure we find that this morphism is an isomorphism of algebraic spaces. □